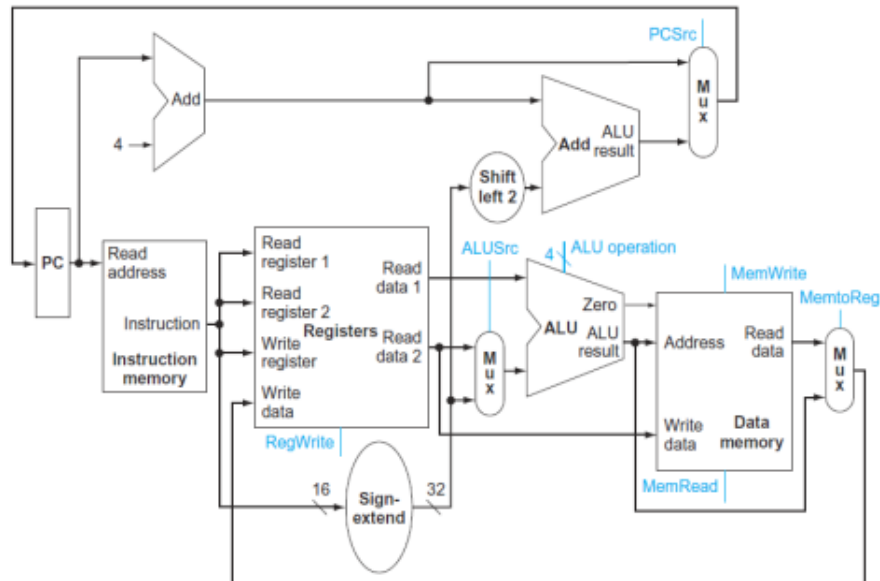


Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/24)

- Consider our simple processor data path shown below (COD Figure 4.11). Assume the delays of units are as follows: PC (read): 10 ps, PC (write): 10 ps, I-Mem (read): 175 ps, Adder: 70 ps, Mux: 20 ps, ALU: 95 ps, RegFile (read): 70 ps, RegFile (write): 70 ps, SignExtend: 10 ps, Shift: 20 ps, D-Mem (read): 250 ps, D-Mem (write): 250 ps.



- State which components are on the **critical path** for a MIPS **load** (lw) instruction.

Answer:

The longest path is to read PC, read the instruction from I-Mem, read RegFile, compute the address via the ALU, read D-Mem, use MemtoReg mux and write RegFile.

Note that the SignExtend and ALUSrc mux are off the critical path for a load instruction as that path is resolved before the RegFile(read) is for ALU operand 1.

- What is the corresponding critical path **delay** for a MIPS **load** (lw) instruction?

Answer:

The critical path delay is $10 + 175 + 70 + 95 + 250 + 20 + 70 = 690$ ps.

